GGAMBOA SAURY Pablo David
NNGUYEN Phan Xuan Anh

GRAPHITE-BASED STRAIN SENSOR ON PAPER

MAIN FEATURES

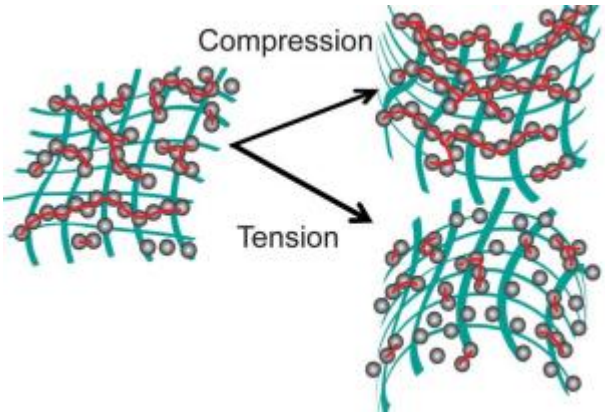
- Low cost
- Small size
- High sensitivity
- Easy production

GENERAL DESCRIPTION

A low-cost, easily reproducible strain sensor fabricated by depositing (drawing) a graphite trace onto porous paper. The device is lightweight, flexible, and suitable for rapid prototyping and low-resource applications.

WORKING PRINCIPLE

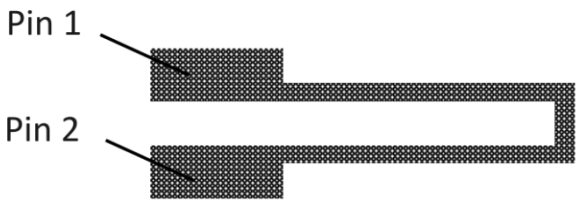
The pencil lead is mechanically deposited into the porous paper substrate, forming a conductive trace composed of graphite particles dispersed in an insulating clay binder. This creates a percolating conductive network embedded within the paper fibres. Mechanical strain alters network connectivity—reducing inter-particle contacts under tension and increasing them under compression—resulting in measurable changes in electrical resistance.



PIN DESCRIPTION

Pin number	Pin connection
1	Supply Voltage
2	Output Singal

Both pins are invertible



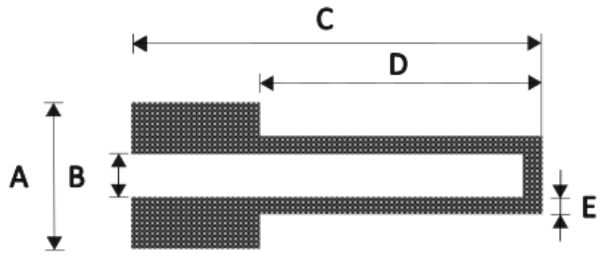
SPECIFICATIONS

Sensor type	Supply (+5V)
Measurand	Resistance
Output signal type	Analog
Power supply	5V
Substrate material	Paper
Graphite types	Graphite deposited from a pencil (HB, 2B, 4B, 6B)

Substrate thickness	0.2

Dimensions

Dimension		Unit
A		mm
B		
C		
D		
E		



Performance Characteristics

Sensitivity

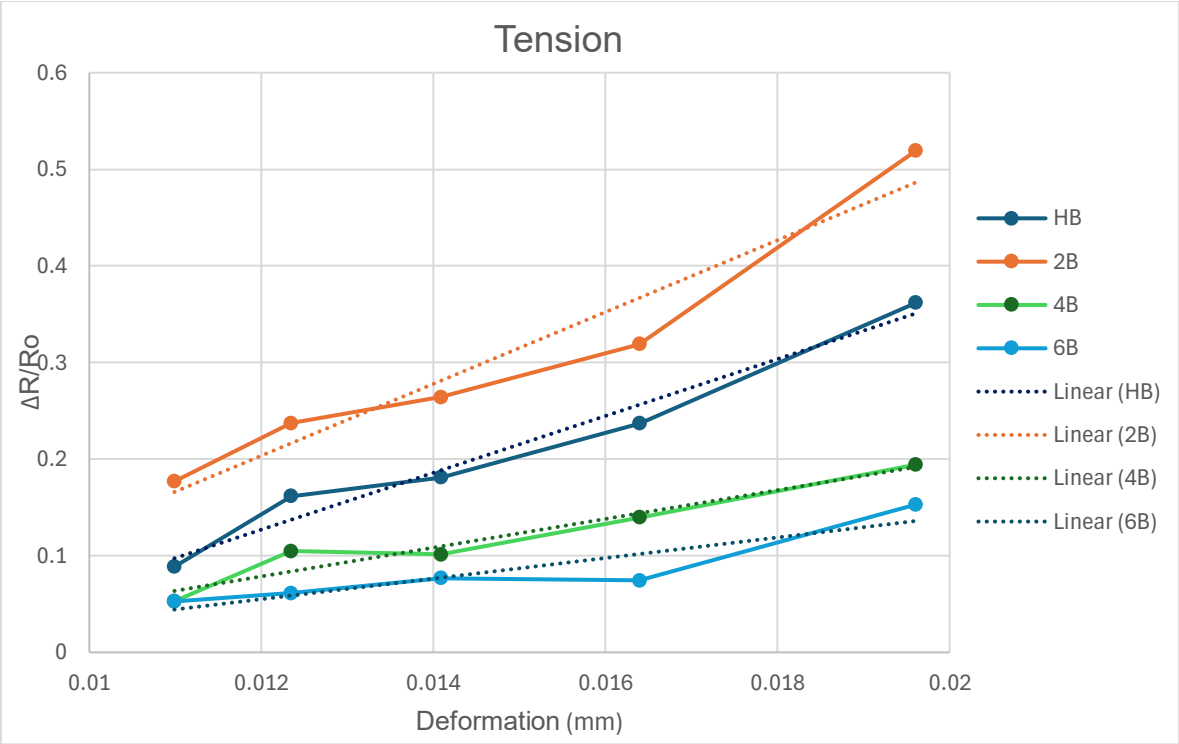
Sensor model	Sensibility (Slope) $\frac{\Delta R/\ R_0}{\varepsilon}$	Linear Fit (R^2)
Tension		
HB	30,812	0,9826
2B	33,63	0,9836
4B	22,069	0,9904
6B	23,46	0,9983
Compression		
HB	-5,8334	0,9963
2B	-12,635	0,9629
4B	-8,7708	0,995
6B	-14,027	0,796

This table´s sensitivity is calculated through linear regression of the measured tange; not adjusted to intercept the zero strain, zero change in resistivity value.

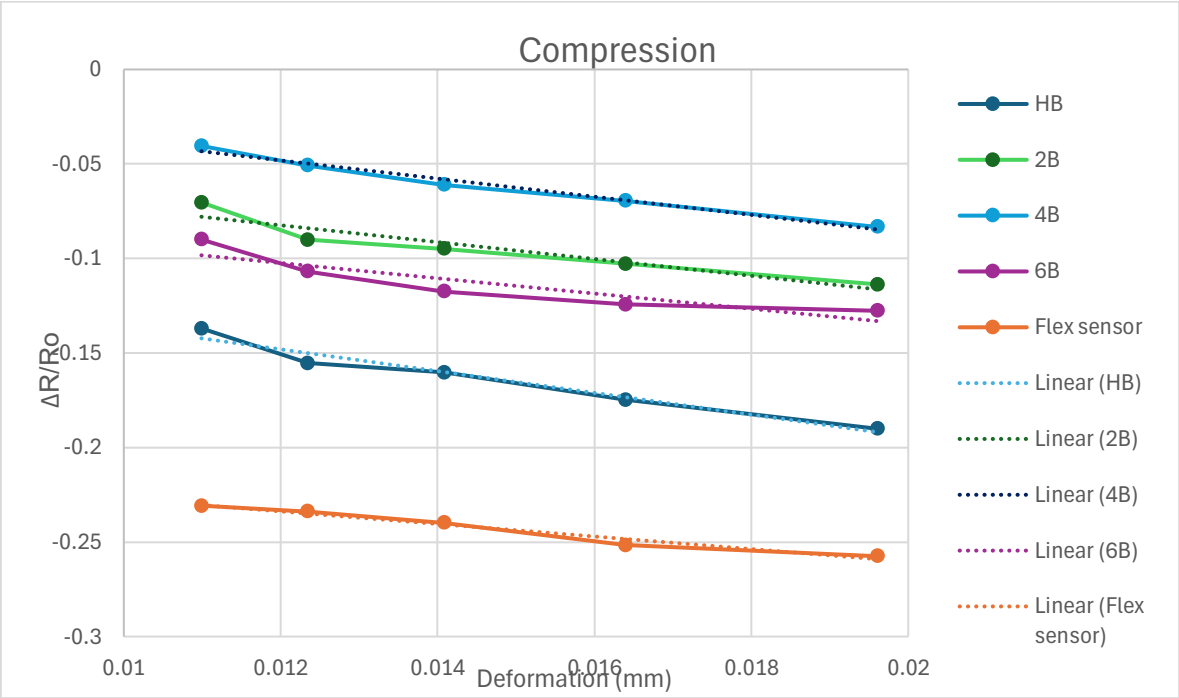
Sensor model	Sensibility (Slope) $\frac{\Delta R/\ R_0}{\varepsilon}$	Linear Fit (R^2)
Tension		
HB	32,958	0,9992
2B	25,014	0,9937
4B	20,295	0,9992
6B	18,365	0,9965
Compression		

HB	-44,673	0,9683
2B	-25,812	0,9885
4B	-11,175	0,9979
6B	-10,792	0,9786

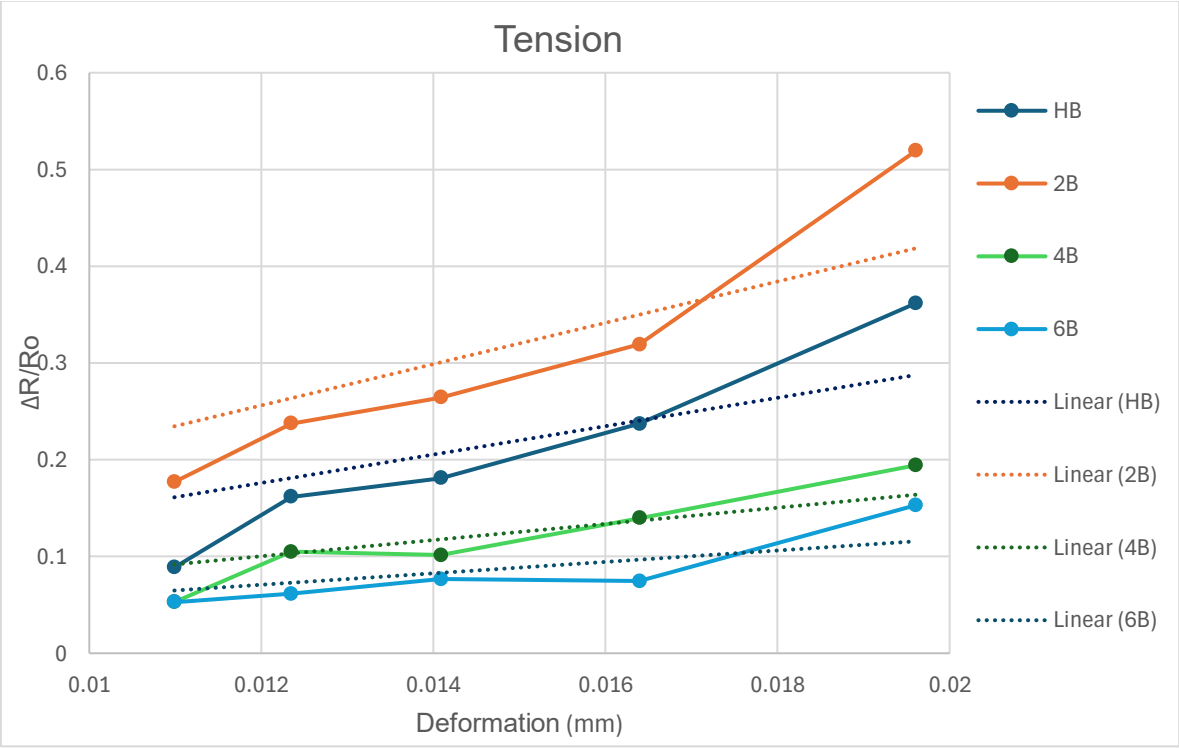
This table's sensitivity is calculated through a linear regression considering the zero strain, zero change in resistivity point. This calculation ignores unknown possibly non-linear behaviour on the non-measured range



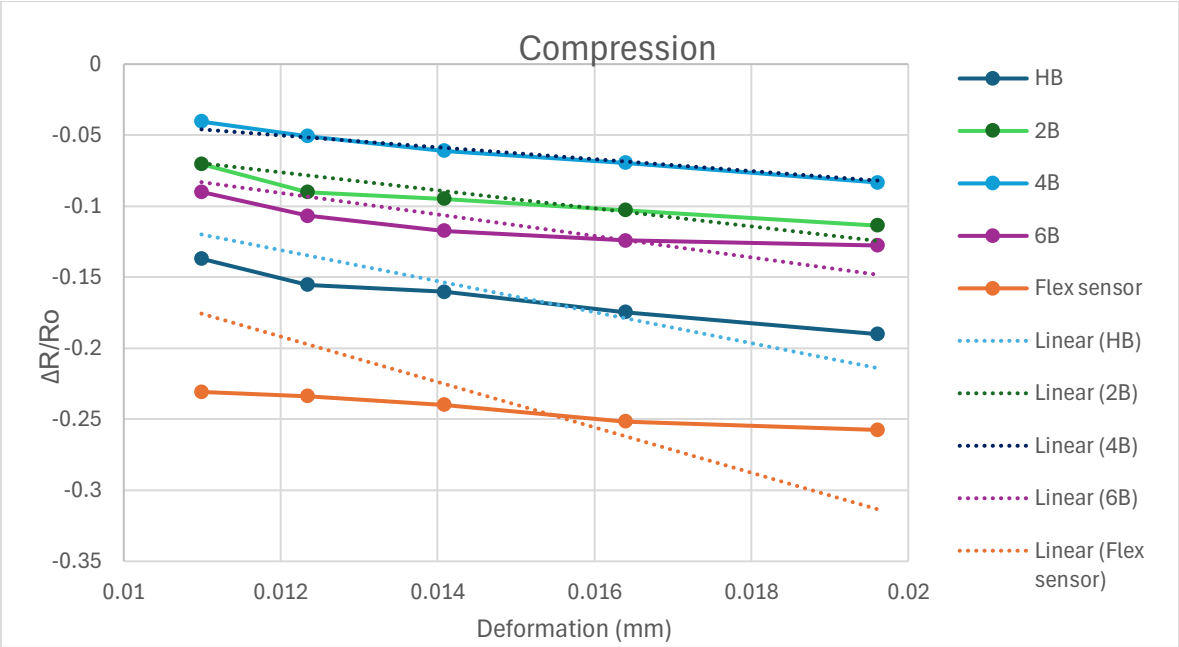
Graph 1. Electrical response in tension of sensor models and commercial flex sensor; linear regression for measured range



Graph 2. Electrical response in compression of sensor models and commercial flex sensor; linear regression for measured range



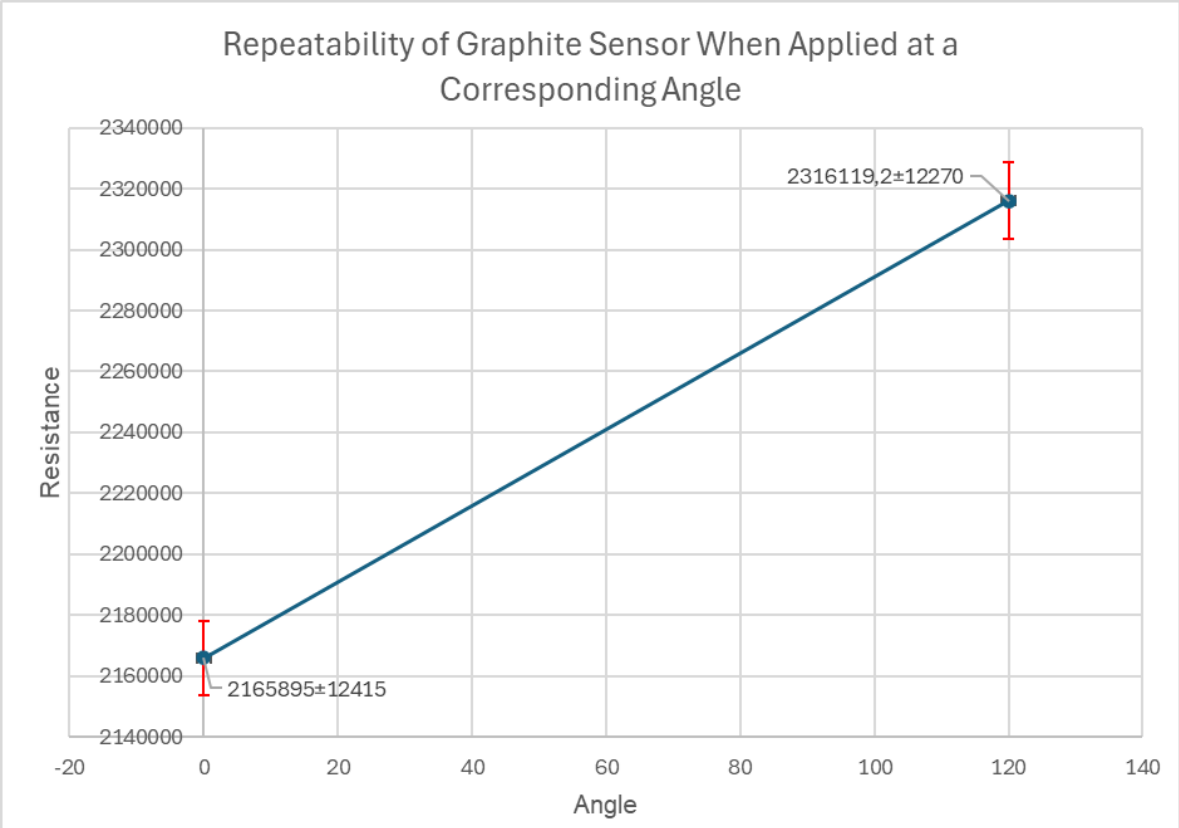
Graph 3. Electrical response in tension of sensor models and commercial flex sensor; linear regression considering the zero strain, zero change in resistivity point.



Graph 4. Electrical response in compression of sensor models and commercial flex sensor; linear regression considering the zero strain, zero change in resistivity point

REPEATABILITY

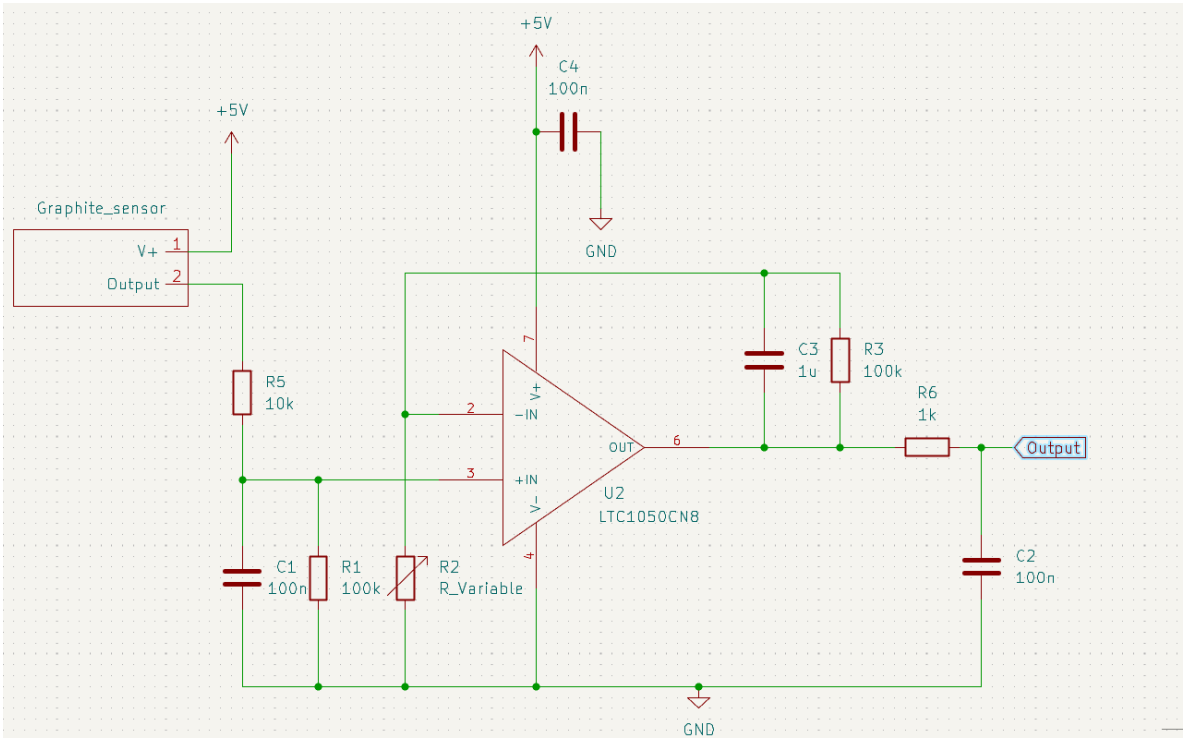
A servomotor was used to apply angles of 0 and 120 degrees successively to the 2B graphite sensor, and the corresponding electrical resistance was measured. The experiment was repeated multiple times to investigate repeatability. In this graph, the value corresponding to each angle is the average of all resistance values measured at that angle. The error bars are derived from the calculation of standard deviation.



GGAMBOA SAURY Pablo David
NNGUYEN Phan Xuan Anh

TYPICAL APPLICATION

A typical application of the sensor is shown in the analog circuit diagram, which uses a transimpedance amplifier to convert and amplify an input current signal into a voltage output signal.



The resistance of the graphite sensor can be calculated through the following relation:

$$R_{\text{Graphite sensor}} = \left(1 + \frac{R_3}{R_2}\right) \times R_1 \times \frac{V_{CC}}{V_{ADC}} - R_1 - R_5$$

Response characteristics of commercially available «flex sensor»

Sensor model	Sensibility (Slope) $\frac{\Delta R / R_0}{\varepsilon}$	Linear Fit (R^2)
Tension		
Flex sensor	38.875	0.9933
Compression		
Flex sensor	-3.3202	0.9712

This tables presents the linear regression without the zero intercept

Sensor model	Sensibility (Slope) $\frac{\Delta R / R_0}{\varepsilon}$	Linear Fit (R^2)
Tension		
Flex sensor	98.122	0.9802
Compression		
Flex sensor	-15.98	0.9735

This tables presents the linear regression with the zero intercept